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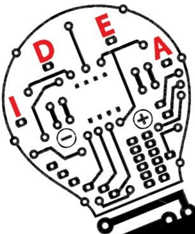

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TABLE I
EFFECTS OF RESISTANCE

Resistance (ohms)	Ipeak (amp)				
Cycle no.	1	10	20	40	50
0	870	870	870	870	870
0.5	870	771	685	557	507
0.9 (original value)	870	706	582	423	368
1.5	870	624	469	300	249
2.5	870	518	345	190	150

into the effect of various transformer parameters on inrush current. Some of these effects are described below.

Effects of system resistance. The effects of system resistance can be considered by varying the value of resistance while keeping other parameters constant. It can be seen that increasing the resistance causes faster decay in the amplitude of inrush current. Therefore it can be said that transformers located closer to the generating plants have a small value of system resistance and thus display higher amount of inrush currents lasting much longer than the transformer installed electrically away from the generator (as they have higher system resistance).

This effect can be analysed in the above testing example by varying the value of source resistance while keeping all other parameters constant. The results are shown in Table I.

Analysis of results. Current does not decay when resistance is zero. This clearly shows that inrush current decays because of damping produced by system resistance.

As the overall resistance increases due to increase in system resistance (transformer winding resistance is constant), inrush current decays faster. This can be observed by comparing the value of inrush current at different cycles for various values of resistance.

Effects of the core material. The effect of core material and thus the effect of residual flux density (Br) can be analysed in the above example by selecting different core materials while keeping all other parameters constant. The results are shown in Table II.

Analysis of results. As is seen from Table II, the first-cycle peak current changes significantly when the residual flux varies (with change in core material). Also the results indicate that switching at $\theta = 90°$ may not necessarily reduce the magnitude of inrush current. So, for reducing inrush current, an appropriate switching angle by considering residual flux must be selected.

TABLE II
EFFECTS OF CORE MATERIAL

Switching angle (degrees)	Ipeak (amp) {first-cycle peak}	
Material	Cold Rolled Steel	Hot Rolled Steel
0 deg	858.36	751.46
30 deg	786.75	679.85
60 deg	591.1	484.2
90 deg	323.85	216.95
120 deg	56.6	-50.31
150 deg	-139.05	-245.95
180 deg	-210.66	-317.56

Applications

This software can be very useful for design engineers, practising engineers and electrical engineering students to study and analyse the behaviour of inrush current under different operating conditions and for different transformer parameters. For example, design engineers can select appropriate values of peak flux density, core material, saturation flux density and transformer dimensions to estimate the value of inrush current, effect of switching angle and harmonic effects before transformer manufacturing and thus can optimise the transformer design.

EFY note. Mathematical models for inrush current and harmonic analysis fundamentals along with relevant mathematical equations, as well the source code of this article are included in this month's EFY DVD and are also freely available for download at *source.efymag.com.* EFY

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